

Project Name: IRON

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Collaborators: Aliyah Zaizay, Nathaniel Boateng

Project Description:

IRON will be a workout tracker that will be able to track the user's progress and gain overtime while giving suggestions for progress based on the user's goals. Suggestions will be based on the user's effort while performing a workout, which will be tracked by using reps in reserve (RIR), adjusting weight and reps within the user's own rep ranges rather than replacing them, with exercises tracked per-machine so history stays accurate across different gyms. As the project advances we will try to develop a food/calorie tracking system where users can accurately track their weight as well as a form reviewer, where the user can record workouts and the app will give you judgment based on how your form looks. Those advanced features will most likely be added on after the semester ends.

Semester Goal:

The goal for this semester is to build a fully functional mobile workout tracker where users can:

- Create and manage workout splits
- Log exercises, sets, weight, reps, and RIR
- Track exercise history by machine/equipment
- Follow a rotating workout split
- View strength progression over time for each exercise and each muscle, including estimated 1RM

The application will use a live Supabase/PostgreSQL backend. A research-grounded RIR progression and feedback engine is a **stretch goal**, dependent on first completing the core tracker and generating real workout data.

Tech Stack: React Native (Expo) · Python (FastAPI) · Supabase (PostgreSQL) · GitHub

Timeline & Responsibilities:

Olakiite - Backend & Progression Logic

- Peer-reviewed research library compiled (RIR, volume, load, 1RM formula); Supabase schema live with auth and one working end-to-end read/write path **by Week 3**
- Core logging flow complete: exercises, sets, per-machine history, split-day rotation, all hitting Supabase directly **by Mid-semester (mid-October)**
- Progression trend display (estimated 1RM, charts) live; RIR-based feedback engine prototype in FastAPI, grounded in the compiled research **by End of November**

Aliyah

- Week 2: Scaffold Expo application, navigation, and core screen structure.
- Mid-Semester: Complete workout/day builder, workout logging UI, machine selection, and theming.
- End of November: Build progression/history views and integrate live backend data.
- Early December: Complete UI polish, testing, and an installable TestFlight / Play internal testing build.

Nathaniel

- Supabase schema live (tables, relationships, auth) and FastAPI service scaffolded - **Week 3**

- Core CRUD endpoints complete: exercises, sets, per-machine history, split-day rotation, all reachable from the app - **Mid-semester (mid-October)**
- Integration testing across app/Suvpabase/FastAPI, plus contributor documentation for the repo - **End of November**

Final Deliverable

By the end of the semester, a user should be able to complete the full flow:

Account → Clarify Goals → Create Split → Start Workout → Log Sets/RIR → Save Workout → View Machine-Specific History → View Progression